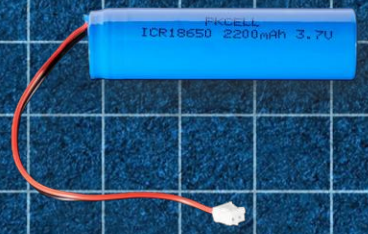
# PARKING COUNTER SYSTEM

AZWAD K. ZAHRAAN

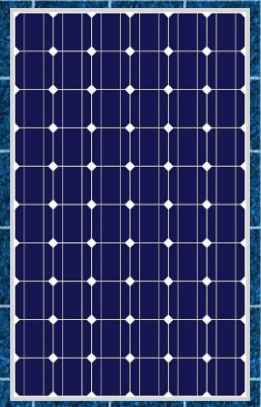
Senior Capstone Project



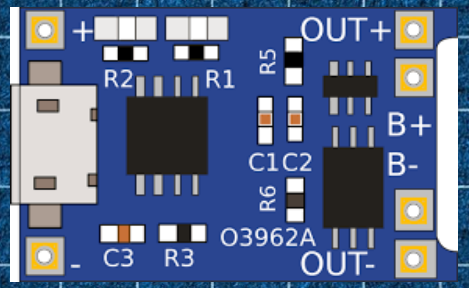
# STARTING COMPONENTS



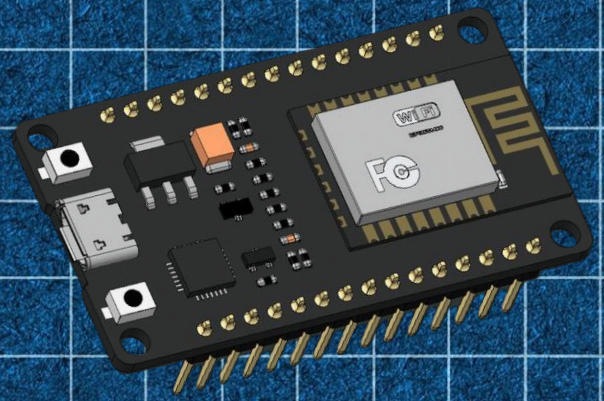

Rechargeable Battery



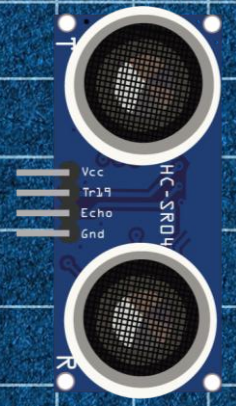
Solar Panel




Charging Module

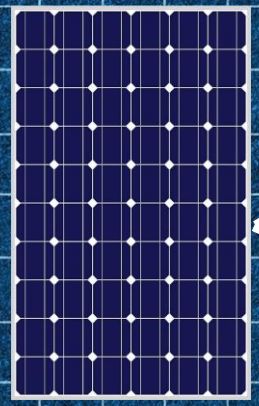


ESP32 E Dev C



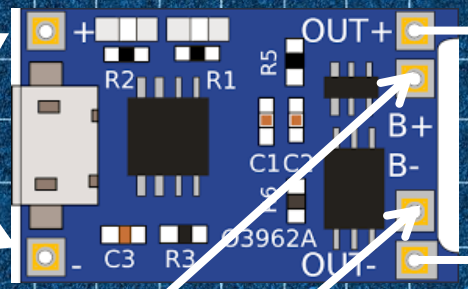

HC-SR04
Ultrasonic Sensor

# PROTOTYPE 3.0



Solar Panel

Charging Module




+(Positive)

3.7 V

- (Negative)



Boost Converter

Parallel Connection

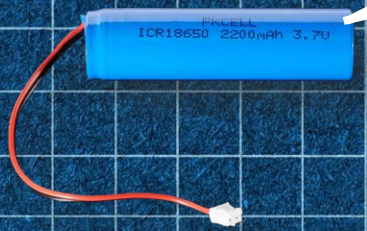
5 V

(+)

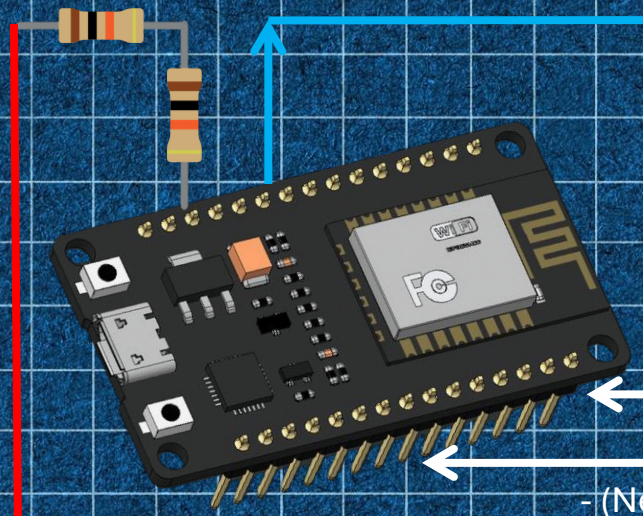
(--)



10 Uf



18650 Lithium-Ion Rechargeable Battery



ESP 32 Dev Kit V4



470 Uf

+(Positive)

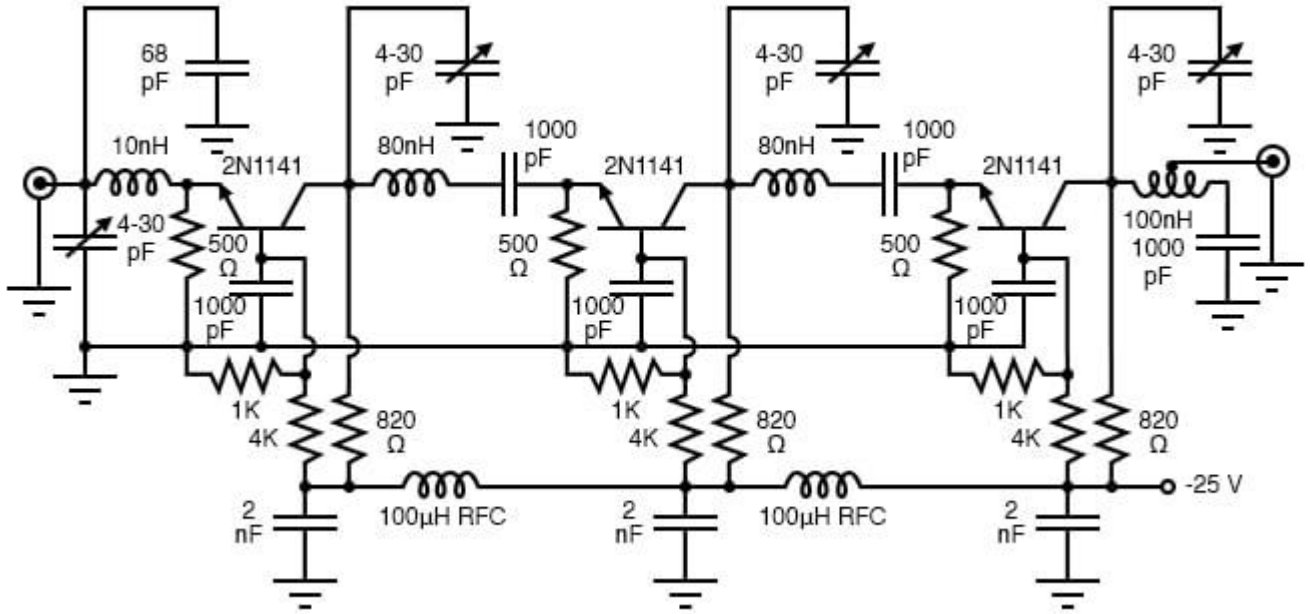
- (Negative)




HC-SR04 Ultrasonic Sensor

# Challenges Overview

- Cable Failure and Port Issues
- Using Multimeter and Soldering Ki
- Electrical Circuits and Connectivi
- Reading Electrical Schematics
- Understanding Network IP Connections across multiple IOT Devices
- How to use Capacitors + Resistors effectively
- **How to affordably solve problems!**




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Kansas Small Business Development Center
Koch Center for Leadership and Ethics
Richel Learning Space
Information Technology
Help Desk

# Thank you